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Fall 2022

Q 1

Find the natural domain of the function, $f(x) = \sqrt{x^2 - 5x + 6}$?

Given.

$$f(x) = \sqrt{x^2 - 5x + 6}$$

Let suppose

$$x^2 - 5x + 6 \geq 0.$$
$$x(x-2) - 3(x-2) \geq 0.$$
$$(x-2)(x-3) \geq 0.$$
$$x - 2 \geq 0 \quad ; \quad x \geq 3$$
$$x \geq 2 \quad ; \quad x \geq 3.$$

Thus the domain of funtion is $(-\infty, 2) \cup (3, \infty)$.

Q 2

Draw the graph of equation $y = 3 - (x+4)^2$?

For x intercept put $y = 0$.

$$0 = 3 - (x+4)^2$$
$$\cancel{0 = 3 - x^2 + 8x + 16}$$
$$3 = (x+4)^2$$

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$x+4 = \pm\sqrt{3}$

$x = -4 \pm \sqrt{3}$.

Now we have two.

x-intercept $(-4+\sqrt{3}, 0)$ and $(-4-\sqrt{3}, 0)$

For y intercept $x=0$.

$y(0) = 3-(0+(4))^2$.

$y = -13$.

y intercept $(0, -13)$.

Now let find extrema using 2nd derivative test.

$\frac{dy}{dx} = -2(x+4)$

$0 = -2(x+4)$.

$x+4 = 0$

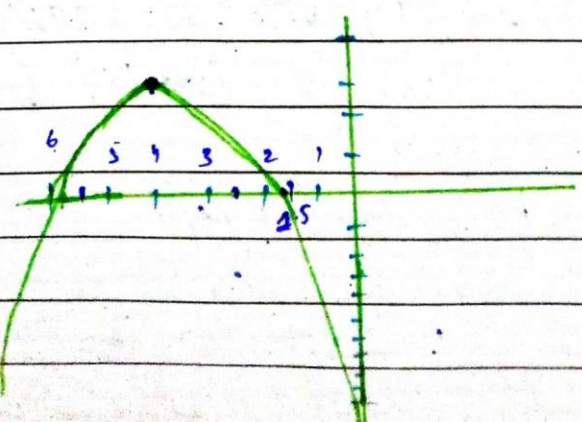
$x = -4$.

Putting $x=-4$ in function.

$y = 3-(4-4)^2$

$y = 3-(0)^2 = 3$.

Now lets draw the graph.



Q3

Find all critical points of $f(x) = x^4(e^x - 3)$?

$$f'(x) = 4x^3(e^x - 3) + x^4 e^x$$

$$= x^3(4(e^x - 3) + x e^x).$$

$x^3 = 0$; $4e^x - 12 + xe^x = 0$.

$x = 0$; $e^x(4+x) - 12 = 0$.

$$e^x(4+x) = 12.$$

$e^x = 12$; $4 + x = 12$.

$e^x = 12$ $x = 8$.

$\ln e^x = \ln 12$

$x \approx 1.07$

$x = 1$

So we have.

($x = 0$; $x \approx 1$; $x = 8$.

Q4.

Find the two intercept of the function $f(x) = x^2 - 5x + 4$ and confirm that $f'(c) = 0$ at same point c b/w those interexcepts.

$$f(x) = x^2 - 5x + 4.$$
$$f'(x) = 2x - 5$$
$$0 = 2x - 5$$
$$x = \frac{5}{2}.$$

$$\cancel{f'(x) = 2x}$$
$$f(x) = x^2 - 5x + 4 = 0$$
$$(x-4)(x-1) = 0.$$
$$x = 4 \;;\; x = 1$$

As $\frac{5}{2}$ lies between $x=1$ and $x=4$ and at this point $f'(c)=0$ so $\frac{5}{2}$ is required point.

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Q.5

Evaluate the integral $\int \frac{\sin\frac{1}{x}dx}{3x^2}$?

$$= \frac{1}{3}\int \frac{\sin\left(\frac{1}{x}\right)}{x^2}dx$$

let $U = \frac{1}{x}$.

$dU = (-1)x^{-2}dx$.

$dU = -\frac{1}{x^2}dx \Rightarrow -dU = \frac{1}{x^2}dx$.

$$= \frac{1}{3}\int -\sin(U)\,dU$$

$$= \frac{1}{3}\int -(-\cos U)\,dU$$

$$= \frac{1}{3}\cos\left(\frac{1}{x}\right) + C.$$

Q.6

Given the function. Compute the limit of the function $y = -2$?

$$g(y) = \begin{cases} y^2+5 & \text{if } y < -2 \\ 1-3y & \text{if } y \geq -2. \end{cases}$$

$\lim_{y\to 2} g(y) = y^2+5 \Rightarrow g(-2) = (-2)^2+5 = 4+5$
$= 9.$

$\lim_{y\to 2^+} g(y) = 1-3(-2) = 1+6 = 7.$

$\lim_{y\to 2^-} g(y) = y^2+5 \Rightarrow g(-2) = 4+5 = 9.$

limit does not exist bcuz $\lim_{y\to 2^+} \neq \lim_{y\to 2^-}$

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Long questions.

Q2

(a).

$f'(x) = ?$ using chain rule
when $f(x) = [x + \csc(x^3 + 3)]^{-3}$?

let $U = x^3 + 3$
and $y = x + \csc U$ then $f(x) = y^{-3}$.

$\frac{dU}{dx} = 3x^2$; $\frac{dy}{dx} = \frac{dx}{dx} - \csc U \cot U \frac{dU}{dx}$.

$\frac{dy}{dx} = \left(\frac{dx}{dx} - \csc(x^3+3)(x^3+3)\frac{dU}{dx}\right)$

$= (1 - \csc(x^3+3)(x^3+3)3x^2$

$\frac{d}{dx}(f(x)) = y^{-3}$

$= -3y^{-4}\frac{dy}{dx}$

$= -3y^{-4}(1 - \csc(x^3+3)(x^3+3)3x^2.$

$= -3[x + \csc(x^3+3)]^{-3}(1 - \csc(x^3+3))$
$(1 - \csc(x^3+3)(\cot(x^3+3)3x^2)$

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(b)

$$f(x) = \begin{cases} x^2+1 & x \le 1 \\ 2x & x > 1 \end{cases}$$

$\lim_{x\to 1} = (1)^2+1 = 1+1 = 2.$

$\lim_{x\to 1} 2x = \lim_{x\to 1} x^2+1$

$2(1) = (1)^2+1$

$2 = 2$

Hence function is continous.

As limit also exist hence function is differentiable.

Q 3

Use any method to find the relative extrema of the function $f(x) = |3x - x^2|$.

$$f(x) = \begin{cases} 3x - x^2 & \text{if } \cancel{x^2}\ \ge 0 \\ -3x + x^2 & \text{if } x < 0. \end{cases}$$

1. For $x \ge 0$.

$f(x) = 3x - x^2$;

$f'(x) = 3 - 2x = 0$;

$x = +\frac{3}{2}$;

2. For $x < 0$.

$f(x) = -3x + x^2$

$f'(x) = -3 + 2x = 0.$

$x = \frac{3}{2}.$

At $\frac{3}{2}$ the behaviour of function change (i.e Negative to positive). ~~so the~~

Substivating a noumber before and after $\frac{3}{2}$ (i.e + in first derivative.

$-3(1.4)+(1.4)^2$; $3(1.6)+-(1.6)^2$.
$=-4.2+1.96$; $+4.8-2.56$.
$=-2.24$; 2.24.

Hence relative extreama is $\frac{3}{2}$.

$\frac{3}{2}$

(b)

$L_1 : x=1+2t$. $L_2 : x=9+t$.
$y=2-t$. $y=5+3t$.
$z=4-2t$ $z=-4-t$.

If both are intersect each other then they should have a come point somewhere amoung them which in this $(7,-1,-2)$.
But we have to verify
$t_1+9=7$ in L_1 and $2t_2+1=7$ in L_2.
$t_1+9-7=0$ $2t_2=7-1$
$t=-2$. $t=3$.

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$-1-2=-t$; $3t+5=-1$

$-3=-t$; $3t=-5-1$

$t=3$; $t=-2.$

and.

$-2=4-2t$; $z=-4-(-2)$

$-2t=-4-2$; $z=-4+2$

$-2t=-6$; $z=-2.$

$t=-2$;

Hence proved.

Q4

Sketch the region enclosed by the curve $y=2+|x-1|$ and $y=-\frac{1}{5}x+7$ and find its area using definite integral.

$y=2+|x-1|$

$y=2+x-1$; $y=2-x+1$

$y=1+x$; $=3-x.$

$1+x=-\frac{1}{5}x+7.$; $3-x=-\frac{1}{5}x+7.$

$x+\frac{1}{5}x=7-1$; $3-7=\left(\frac{-1+5}{5}\right)x$

$\frac{5+1}{5}x=6$;

; $-4=\frac{+4}{5}x.$

$\frac{6x}{5}=6$;

; $x=-5$

$x=5$;

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let evaluate Area:

$$\int_{-5}^{5} \left(2+(x-1)+\frac{1}{5}(x-7)\right) dx$$

$$= \int_{1}^{5} \left(2-(x-1)+\frac{1}{5}(x-7)\right) dx + \int_{-5}^{1} \left(2+(x-1)+\frac{1}{5}(x-7)\right) dx$$

$$= \left[-\frac{3}{5}x^2+6x\right]_1^5 - \int_{-5}^{1}\left(\frac{6}{5}x-4\right) dx$$

$$= \left[-\frac{3}{5}x^2+6x\right]_1^5 + \left[\frac{3}{5}x^2-4x\right]_{-5}^{1}$$

$$= \left[\frac{3}{5}(5)^2-6(5)\right] + \left[\frac{3}{5}(1)^2+6(1)\right]$$

$$- \left(\left[\frac{3}{5}(1)^2-4(1)\right] + \left[\frac{3}{5}(-5)^2-4(-5)\right]\right)$$

$$= \left[-15+30\right] - \left[\frac{3}{5}(1)+6\right] + \left[\frac{3}{5}-4\right] + \left[15+20\right]$$

$$= \left[+15-5.4\right] - \left[\frac{3-20}{5}-35\right]$$

$$= 9.6 + \frac{(-192)}{5}$$

$$= 29.4.$$